

Network Science HS23

Assignment 4

Blockchain & Distributed Ledger Technologies Group

FOR STUDIES PURPOSES ONLY

Instructions:

1. You must submit your answers through OLAT.
2. The deadline for submission is 22.12.2023 at 23.59.
3. Answer means
 - Code or theoretical solution, depending on the exercise request.
 - A document with the results clearly visible.
 - Possible formats are PDF, Python Notebooks.
 - All code must be already executed and results clearly visible.
4. Show all your work and reasoning clearly.
5. You can do this assignment in groups of up to two.

1 Compartmental Models on Networks

1. **Task:** Write a function to simulate the SI dynamic on a generic network and provide a visualisation.

The function should take in input:

- a network, e.g. if you are using `networkx` a generic `nx.Graph`,
- the infection rate β ,
- an infected seed node.

You may also want to add other parameters, e.g. the maximum number of steps for the simulation (some bad things may happen if you don't). As an output, the function should return the list of nodes infected in the simulation. Additionally, run the SI function for an adequate number of steps, with $\beta = 0.5$ on the following two graphs (the datasets are provided), with infected seed node "10":

- `graph1.1.gml`
- `graph1.2.gml`

and provide a visualisation of the final state of the epidemic, where all the infected nodes at the end of the run are red, or blue if they are not infected.

2. **Task:** To extend the function you wrote for point 1 in order to simulate the SIS model. You need to introduce γ , i.e. the recovery rate, to the input parameters of the function.

Provide a plot of ρ^I and ρ^S as a function of the simulation time-step, on graph `graph1.1.gml` with $\beta = 0.2$ and $\gamma = 0.4$ over 1000 steps (at least). Choose a random seed such that you don't end up in the trivial stationary states $(\rho^I, \rho^S) = (0, 1)$

Hint: in order to observe "smoother" result you can use a moving average. You can check out moving average functions on `pandas` python module, or implement it yourself.

3. **Task:** Observe how the infected share of the population stationary state is affected by the network topology, in the SIS model. You have to plot the stationary state share of infected nodes $\rho^I(t = \infty)$ as a function of β/γ , on two different kind of network structures:

- For the Erdos-Renyi $N = 500$ and $p = 6/N$
- For the Barabasi-Albert $N = 500$ and $m = 3$

You can use `networkx` native functions to create the networks. You need to average ρ^I on multiple SIS runs, a sample size of 100 SIS simulations per β/γ combination will work. For the x-axis of the plot $\frac{\beta}{\gamma}$ chose a logarithmic scale with at least 10 values in the interval $[0, 1]$.

On the y-axis you need to plot:

- $\rho_{ER}^I(t = \infty)$, the share of infected nodes in the stationary state for the Erdos-Renyi network, averaged over at least 100 simulations as a function of β/γ
- ER critical epidemic threshold R_C .
- $\rho_{BA}^I(t = \infty)$, the share of infected nodes in the stationary state for the Barabasi-Albert network, averaged over at least 100 simulations as a function of β/γ
- BA critical epidemic threshold R_C .

Hint: It may take a while to run the simulations, take this into account.

4. **Task:** observe the effect of different immunisation strategies. Modify your SIS function to introduce the immunised state, a state where the node cannot get infected nor spread any more. Then implement two different immunised strategies:

- Random strategy: immunised a fraction $g \in [0, 1]$ of the population, where the nodes immunised are chosen randomly.
- Target Strategy: consider a fraction g of nodes randomly. For each node randomly chosen, immunised their largest degree neighbour.

To compare the strategies, run the epidemics simulations on a Barabasi-Albert network with $N = 1000$ and $m = 3$ with $\beta = 0.3$ and $\gamma = 0.6$, immunised with g ranging $\in [0, 1]$. Produce a plot where:

- On the x-axis there is g the share of immunised nodes.
- On the y-axis there is $\rho^I(t = \infty)$, the stationary state of share of infected nodes, as a function of g , for the two immunisation strategies.

Hint: Keep in mind you want to compute the average value of $\rho^I(t = \infty)$ over multiple run, to avoid small sample bias.

Hint: It may take a while to run the simulations, take this into account.

2 Centralities and influence

1. **Task:** Write a function to simulate the SIR dynamic on a generic network and provide visualisation.

The function should take in input:

- a network, e.g. if you are using `networkx` a generic `nx.Graph`,
- the infection rate β ,
- the recovery rate β ,
- an infected seed node.

. For the visualisation, you have to plot each of the requested network, using a continuous colormap to denote each node Q_i . A colorbar for interpretation should be provided.

Graphs to consider:

- A Barabasi-Albert network with $N = 100$ and $m = 3$.
- A 1D regular lattice with periodic condition, with $N = 100$ nodes and coordination number $k = 6$.

Hint: You have to map the Q_i values to a continuous colormap. The colorbar is needed to interpret the plot.

Hint: We defined the SIR Q_i in class. In order to have a reliable result, for each node you need to average Q_i over a large sample of simulations.

2. **Task:** Evaluate the SIR spreading ability of the nodes vs their centrality.

For each of the required network use the SIR function you defined in the previous point to compute the correlation coefficient (Spearman correlation) of the spreading ability Q_i with the following centrality measures: degree, closeness, PageRank and k-core centrality. Plot the correlation coefficient as a function of $\lambda = \beta/\mu$ in the range $[0.25, 4]\lambda_C$, where λ_C is the epidemic threshold, with step size 0.25.

Graphs to consider:

- Erdos-Renyi with $N = 100$ nodes and $p = 6/N$.
- Barabasi-Albert with $N = 100$ nodes and $m = 3$.
- Watts-Strogatz with $N = 100$ nodes, rewiring probability $p_r = 0.02$ and coordination number $k = 6$.

3. **Task:** Produce the scatter plot of centrality measure vs spreading ability. Do a scatter plot of Q_i vs. each centrality measure x_i from the previous task, at the critical point $\lambda = \lambda_C$ by normalizing the maximum value to one as $Q/\max(Q)$ and $x/\max(x)$, respectively. Graphs to consider:

- Erdos-Renyi with $N = 100$ nodes and $p = 6/N$.
- Barabasi-Albert with $N = 100$ nodes and $m = 3$.
- Watts-Strogatz with $N = 100$ nodes, rewiring probability $p_r = 0.02$ and coordination number $k = 6$.

4. Reproduce the scatter plot of point 3 on the real-world networks provided. Comment the results, taking into accounts the underlying network meaning and functionality. Provided datasets:

- `graph_madrid.gml`: A network of associations among the terrorists involved in the 2004 Madrid train bombing, as reconstructed from press stories after the fact Cardillo et al. [2013].

- `graph_jazz_collab.gml`: Jazz collaboration network. Nodes represent jazz musicians and Edges represent collaborations in bands that performed between 1912 and 1940, Gleiser and Danon [2003].

Hint: these calculations may take a lot of time, plan in advance.

References

- Cardillo, A., Gómez-Gardenes, J., Zanin, M., Romance, M., Papo, D., del Pozo, F. and Boccaletti, S. [2013]. Emergence of network features from multiplexity, *Scientific Reports* **3**: Article number: 1344.
- Gleiser, P. M. and Danon, L. [2003]. Community structure in jazz, *Advances in complex systems* **6**(04): 565–573.